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High-resolution Fluid Flow Simulation in X-crossing Rough Fractures

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ABSTRACT: Modeling of fluid flow in rock fractures is a key issue in answering numerous geoengineering problems in the fields of geophysics, reservoir engineering, rock mechanics, to geothermal processes. Although fluid flow in single fractures has been extensively studied in the last 7 decades, fractures commonly seen in fractured reservoir systems intersect each other forming complex geometric structures. Such fracture networks likely affect fluid flow behavior and solute transport. To investigate the impact of geometric characteristics of connected rough-walled fractures with an X junction shape on linear and nonlinear fluid flow behaviors, a sensitivity analysis was carried out by conducting a series of numerical fluid flow simulations on X shape fractures, generated by scanning real rock fractures with distinct roughness (low, medium, and high), intersecting angles, and apertures. The fluid flow through these fractures at different flow rates was simulated by solving Naiver-Stokes equations. The results show that tortures paths and the formation of eddies are more accentuated on the rough fracture than the smoother ones, and the tortuosity of the streamlines is related to the roughness and the geometric characteristics of the intersection. Simulation results of the different models were compared, which show that the intersection significantly impacts the relationship between the hydraulic gradient and the flow. Therefore, the pressure gradient increases with the decrease of the intersection angle especially for low aperture and rough cases.

1. INTRODUCTION

Understanding fluid flow through a fractured rock mass has great importance to numerous underground industrial activities, such as geothermal extraction (Zhao, 2016), CO2 storage (Catherine Noiriel et al., 2013), oil and gas exploration (Bo Li et al., 2016), underground oil storage (Qiao et al., 2017; Wang et al., 2015), and hydraulic fracturing (Blanton TL,1982).

It is important to measure the impact of roughness to improve the performance of large-scale models since most existing large-scale models still rely heavily on simplified smooth parallel-plate models and related models for natural rock fractures with rough walls (Zimmerman et al., 1992; Zimmerman and Bodvarsson, 1996; Ge 1997; Bodin et al., 2007; Zhao et al., 2013; Wang et al., 2015).

The influence of surface roughness on fluid flow behavior can be significant (Li et al., 2019). It has been found that roughness generates turbulence and non-linearity in the fractures (Zou et al 2015, Li et al., 2016). Even more intriguing is the fact that low Reynolds numbers ($Re < 1$) in a fracture can also create non-linearity (Xiong et al., 2020).

The past ten years have seen an in-depth examination of how the roughness of a single fracture affects the flow of fluids and how solutes are transported (Deng et al., 2018; Ni et al., 2018; Xiong and Jiang 2018; Wang et al. 2015, 2018; Li et al. 2014a,b, 2017; Briggs et al., 2017 ;Pastore et al., 2017; Tzelpeis et al., 2015; Develi and Babadagli 2015; Chen et al., 2015; Zhang and Nemcik 2013; Lavrov2013; Neuville et al., 2010; Javadi et al., 2010; Qian et al., 2005; Radilla et al., 2013). However, fractures in nature cross each other forming complex geometries due to roughness, aperture, and contact areas. Transport processes in nature cannot be understood without accounting for the multi-stranded, non-planar network character of fracture systems and the fracture intersections that control fracture connectivity (Viswanathan et al., 2022). In addition, the flow behavior at different fracture crossings is of great interest as it can significantly affect transport through the fracture network (Mourzenko et al., 2002).

By running numerical simulations, it is possible to gain more insight into the way fluids behave through intersecting rough fractures. This information can be used to enhance the design of underground structures like oil and gas wells, where accurate forecasts of rock behavior are essential for maximizing production and reserves.

This paper proposes a computational solution to investigate the effect of surface roughness on fluid flow in fracture networks. Accordingly, twelve fracture models were developed by changing roughness, aperture, and angle intersection for simulating single-phase fluid flow through X-shaped, rough-walled fractures. The measurement of roughness characteristics can be done by scanning real rock fractures (Xiong et al., 2011, Li et al., 2019) and were incorporated herein.

The purpose of this paper is to explore the influence of surface roughness on fluid flow through three-dimensional, intersected natural rock fractures and describe the nature of rough fracture surfaces. It then presents results from a computer simulation study that is designed to mimic the results of actual experiments. In particular, the effects of flow velocity on the distribution of fluids in intersected rough fractures will be discussed. The focus of this paper will be on how the intersection influences the fluid flow through different media, and whether or not it acts as a constriction or opening relative to individual fractures?

2. METHODOLOGY

In this study, we present a high-resolution fluid flow simulation of two orthogonally intersected rough-walled fractures, which is one of the most encountered patterns along with bifurcated (Y) and (Z) fractures (Fig.1).

Fig. 1. Fractures in a rock outcrop highlighting the most common naturally occurring: the blue is the X-shaped fracture, in red the Z fracture, and in green is the bifurcated fracture

The proposed approach combines different scanned fracture surfaces of individual mechanically created fractures, modeling of synthetic fracture, and computational fluid dynamic simulations.

2.1. *Samples Collection and surfaces preparation*

In order to prepare the X-crossed fracture numerical models for the fluid flow simulation, three physical rock samples were prepared, cylindrical in shape, each having a diameter of 38.1 mm, and a length 76.2 mm (Fig.2) with different grain sizes, coarse, medium, and fine.

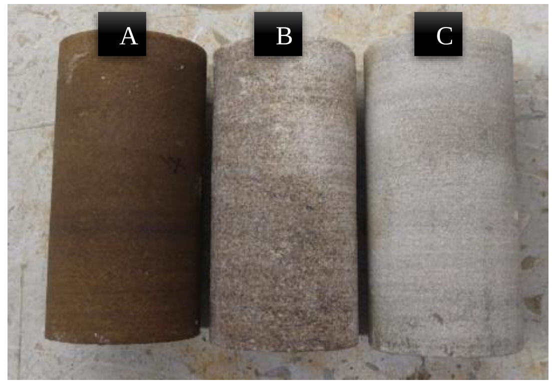


Fig. 2. Specimens used to create rough fracture surfaces. A: fine, B: coarse, and C: medium

All the samples were drilled perpendicular to the bedding in order to eliminate the bedding effect and control the splitting process.

The rough rock fractures were prepared in the laboratory by splitting the intact cylindrical rock samples into two halves (Fig.3) using a compressive testing machine at a rate of 0.1 psi/sec. This created mode 1 fractures

Fig. 3. The experimental setup used to split the core samples.

2.2. *Surface roughness digitization and characterization*

To accurately reproduce the real fracture characteristics, the rock fracture surface was scanned using a noncontact three-dimensional optical high-resolution scanner (Artec Space Spider; Fig.4).

Fig.4. Artec Space Spider 3D scanner used in the study.

The measurement accuracy of this device is 0.05mm for the 3D point accuracy, 0.1 mm for the 3D resolution, and 0.05 mm+0.3 mm/m for the 3D accuracy over distance.

Based on measured data, the three fracture surface was digitized as shown in (Fig. 5). The numerical rock fracture surfaces were used as the primary surfaces to generate the 3D numerical fracture models.

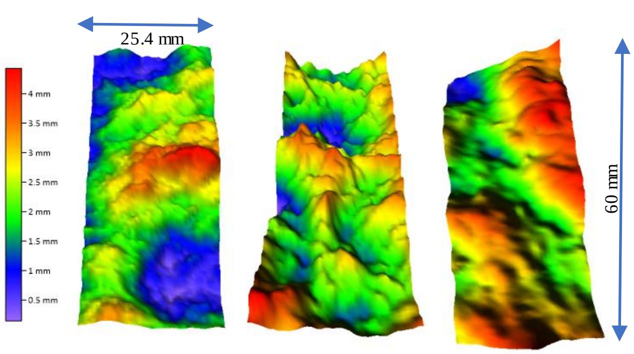


Fig. 5. Scanned rough fracture surfaces, left to right: high, medium, and low roughness.

Regarding the representation of the surface roughness geometrics, the mean Joint Roughness Coefficient (JRC) of 200 cross-sectional profiles in the x direction of each individual rough surface has been estimated (Eq. 1) using a relation developed by Li and Zhang (2015):

$$JRC = 98.718 \cdot Z_2^{1.6833} \qquad \text{(Eq. 1)}$$

With:

$$Z_2 = \left[\frac{1}{N_t}\Sigma\left(\frac{Z_{i-1}-Z_i}{x_{i-1}-x_i}\right)^2\right]^{1/2} \qquad \text{(Eq. 2)}$$

Z2 (Eq. 2) is defined as the root mean square of the first deviation of the 2D profiles along the surface. A Matlab code was used for the estimation of statistical roughness parameters and JRC for 250 profiles of rock fracture surface along the direction parallel to the flow direction. The Z2 profile standard deviation of the three surfaces is summarized in table 1.

Table 1. Z2 standard deviation and mean surface JRC.

surface	Z2 standard deviation	Mean JRC
High roughness	0.017203	~15
Medium roughness	0.033441	~10
Low roughness	0.008467	~5

To make a realistic numerical reproduction of surface roughness in natural rock fractures, featuring contact area, asperities, and complex spatially variable aperture, an anisotropic fracture model with varying aperture sizes and asperity contacts was constructed. A numerical 'uplift' step that raised the scanned surface by 1 mm and 0.7 mm respectively followed by a 'shear' movement that shifted the upper surface along the X direction with a shear displacement of 1.0mm to build the single fracture models.

The single fracture models were then duplicated and rotated along the y-axis to 10° and 90° respectively to make the X-shaped fracture models.

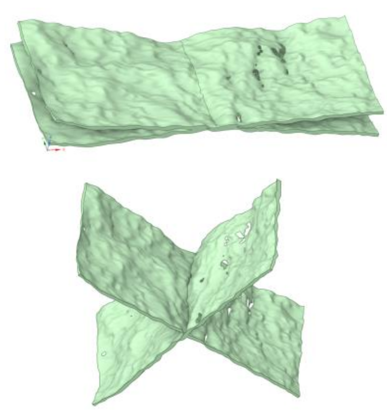


Fig. 6. Examples of 3D X-crossed fracture models JRC=10, intersection angle 10° and 90 °, respectively, (white spots within the green surface elevation models represent places where the adjacent rocks are in contact)

2.3. Meshing

The tetrahedron meshing method was used over the three-dimensional geometric models in order to better fit the fracture replica (Fig. 7). The complexity and the number of tetrahedron elements depended on the fracture roughness and the aperture.

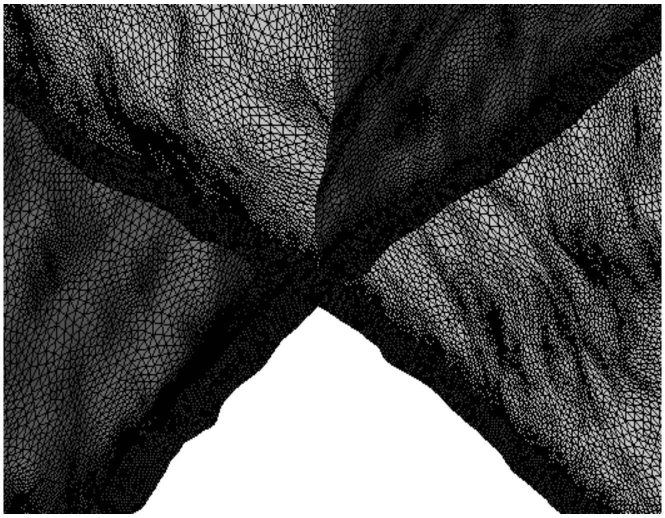


Fig.7. An example of meshing used in numerical simulations (Model: JRC=10, 1mm, 90°).

To ensure mesh-independent results for each model, a large number of mesh elements have been used, between 6 million and 16 million of elements, depending on the aperture and the roughness.

Two mesh metrics are used to assess the mesh quality. the element quality, which is the ratio of the volume to the square root of the cube of the sum of the square of the edge lengths varying from 0 to 1 (Fig. 8), and the aspect ratio, which is a measure of the stretching of a cell, and is defined as the ratio of the maximum value to the minimum value of the normal distances between the cell centroid and face centroids.

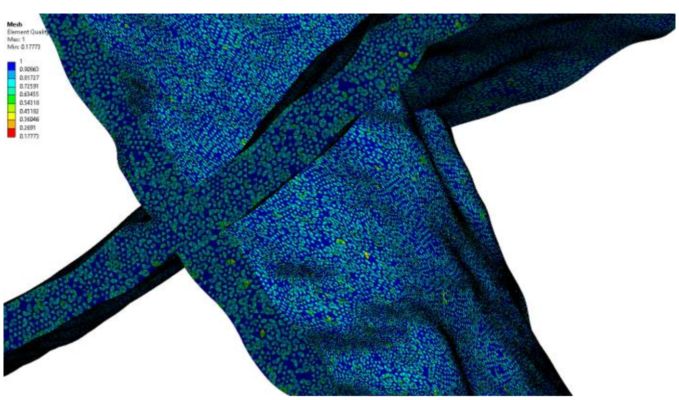


Fig.8. Example of the mesh quality distribution (model: JRC=5 aperture =1mm angle=90°)

In this study, we insured that all the prepared models targeted an aspect ratio close to 1 and 96 % of the elements have above 0.6 element quality (Fig. 9).

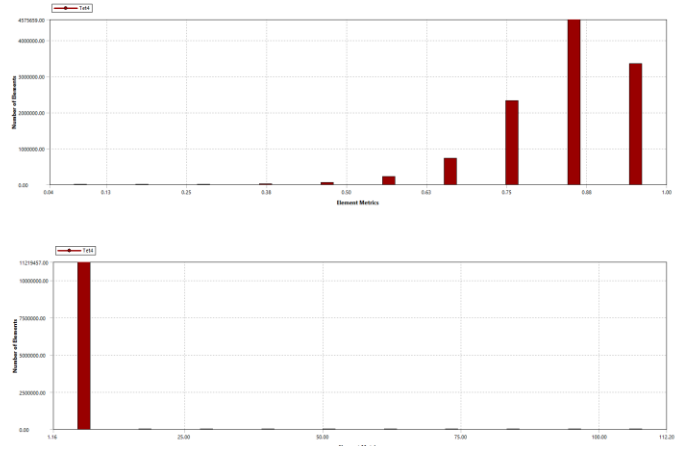


Fig.9. Example of the element quality (top) and the aspect ratio (bottom), model: JRC=10, 1mm and 90°.

Based on the number of elements used in each model, there is no difference in the calculation results. However, the computational time duration increases. As a consequence, the precision of mesh discretization can ensure the accuracy of the calculation results.

2.4. Numerical Simulation

In this study, the commercial Computational Fluid Dynamics (CFD) software, Ansys Multiphysics (Ansys, 2021), and the Fluent module were employed to sequentially solve the Navier Stocks Equation (NSE) (Eq. 3), which is commonly used to describe the Newtonian viscous fluid flow at a microscopic level. Both have been widely employed to analyze the flow of fluids through rock fractures (Huang et al., 2017 Suri et al., 2020 Crandall et all 2010). The NSE is represented by:

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right) + \nabla p = \mu\nabla^2\mathbf{u} + \mathbf{F} \qquad \text{(Eq. 3)}$$

Where the flow velocity vector is given by u, the fluid density by ρ, the pressure gradient of the fluid by ∇p, and the viscosity coefficient by μ.

2.4.1. Initial and Boundary Conditions

Two of the four fractures branches were selected as inlet branches and the remaining two were designated as outlet branches. To facilitate numerical simulations, a constant flow rate was imposed on the two inlet branches rather than a constant pressure. Impermeable non-slip boundary conditions were applied to the fracture walls due to the purpose of the study (Fig.10).

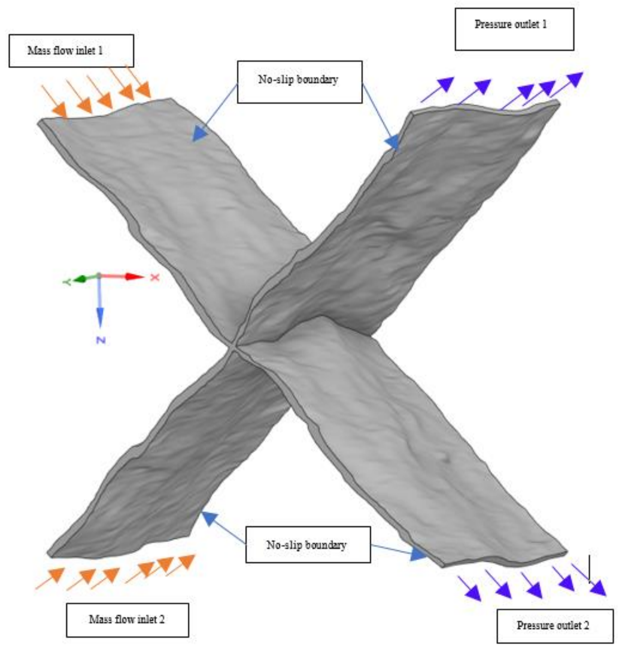


Fig.10. Schematic of the boundary conditions. Example JRC=5 0.7 mm 90°.

Flow rates ranging from 0.1 g/min to 200g/min were simulated with a Reynolds number below 1, which confirms that the flow remains within the linear regime (Zimmerman et al., 2004). This has been further confirmed by analyzing the relationship between fluid rate and hydraulic gradient. Water density and viscosity are 998.2 kg/m^3 and 0.01 poise, respectively.

2.4.2. *Simulation Validation*

For the validation of the model, a simulation sensitivity analysis on four different single fractures (see Fig. 11) was compared to the analytical solution (cubic law; Eq 4) in order to evaluate the accuracy of the proposed simulation model. The geometrical parameters of each fracture are summarized in table 2.

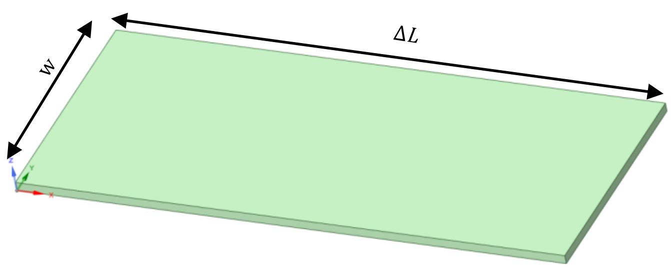


Fig.11 Parallel plate model for validation.

Table 2. Simulation validation models characteristics.

Models	w (m)	e (m)	DL (m)
T1	0.0254	0.0005	0.06
T2	0.0254	0.001	0.06
T3	0.0381	0.0003	0.06
T4	0.0254	0.0005	0.1

The cubic law is expressed as:

$$Q = \frac{we_h^3}{12\mu}\frac{\Delta P}{\Delta L} \qquad \text{(Eq. 4)}$$

Where Q is the flow rate, w is the fracture width, e_h is the hydraulic aperture, μ is the dynamic viscosity, and ΔL is the fracture length.

The result showed that the proposed simulation model is capable of accurately simulating the flow characteristics in the intersecting cracks. The results of the validation four the four different models are shown in Fig.12.

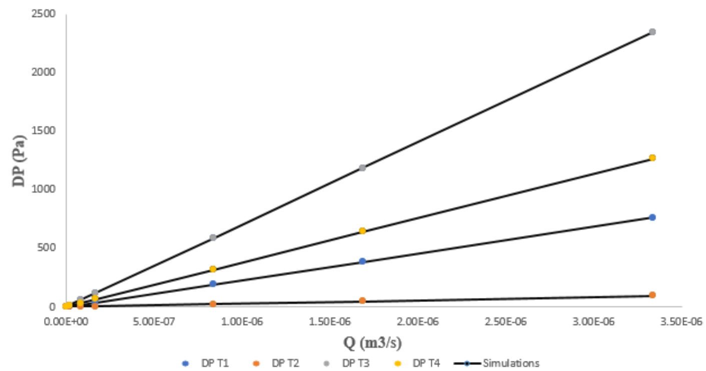


Fig.12. Simulation validation for the four parallel plate models.

3. RESULTS AND DISCUSSIONS

The numerical results in the X-shaped fracture models were taken for illustration and comparison. Fig. 13 shows the 3D distributions of pressure and the streamlines at flow rates of 100 g/min at different roughness degrees and angles. One observation from Fig. 13 is that the pressure varies almost linearly along the flow direction (x-direction) in all the models.

A close look at the pressure drop at the intersections (Fig. 13) indicates a discernible contrast between the 10° and 90° rough models.

In low angle intersection models, it is more probable to observe contact areas and areas with very low aperture. As a result, the behavior of streamlines and pressure drop in these areas may differ significantly from that of high angle intersection models.

Effect of roughness

The flow is maintained in a laminar regime; therefore, the streamlines are constant. The streamlines in the parallel plate models (JRC=0) are uniform and symmetrical no matter the intersection angle (Fig.13), and the fluid coming from inlet-1 is exclusively directed towards outlet-1 (similarly for inlet-2 and outlet-2). However, with an increase in roughness and a decrease in aperture,

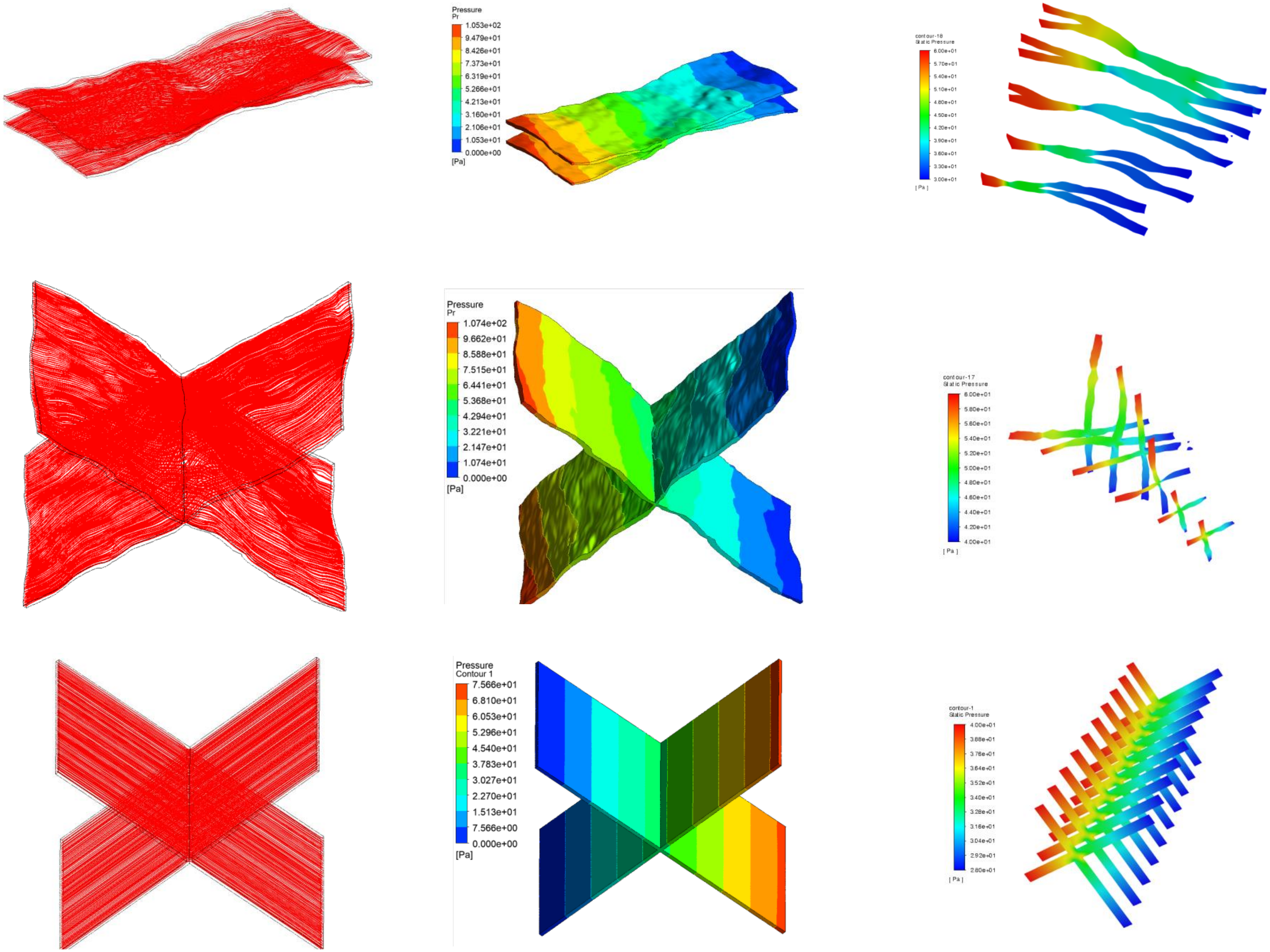


Fig.13 Fluid pressure field for the X-shaped fracture at 0.7 mm of aperture (5 JRC and 0 JRC models) under inlet flow rate of 100 g/min with different intersection angles, 10° and 90°.

some streamlines originating at inlet-1 may be diverted to outlet-2, while simultaneously some streamlines originating at inlet-2 may be diverted to outlet-1 (Fig .14).

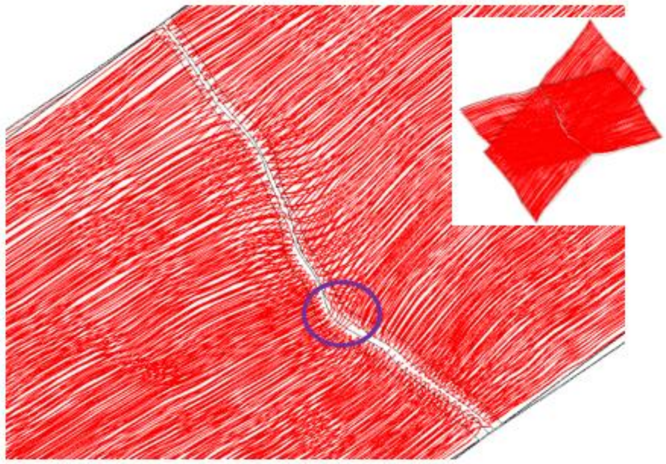


Fig.14. Streamlines behavior for the model 0.7mm, 90°, and JRC=5

The simulation results indicate that the roughness also has an effect on the pressure drop. A linear association between the flow rate and pressure drop was presented by the parallel cross fractures model (JRC=0), however, the non-linear effect increases as the roughness increases (Fig.15).

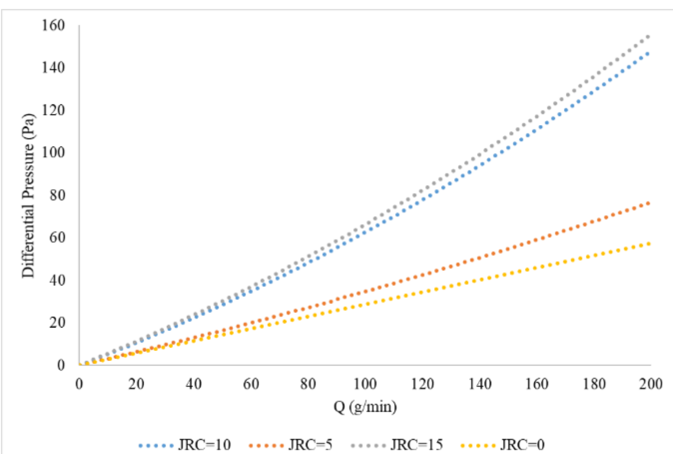
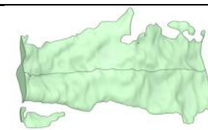
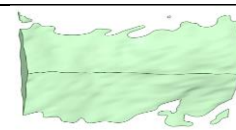
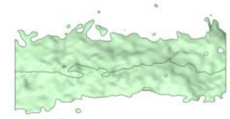
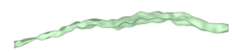
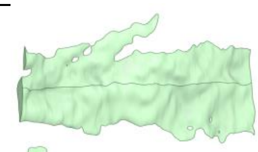
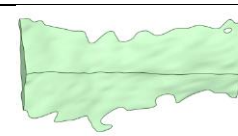


Fig.15. Differential pressure vs flow rate for different roughness (1mm aperture and 90° model).

3.1. *Intersections characteristics and their effect on fluid flow*.

The intersection geometry is related to several factors, for instance, the roughness, the aperture, the shear amount of each fracture, the contact area, and the intersection angle of the two intersected fractures. Table 3. shows the 3D view of the intersections of the three rough-walled models at an intersecting angle of 10° and 90° for different JRC and apertures.

Table 3. 3D view of the intersection at different apertures, roughness and intersection angles.

Aperture	JRC	Angle	Intersection geometry
1 mm	15	10	
		90	
	10	10	
		90	
	5	10	
		90	
0.7 mm	15	10	
		90	
	10	10	
		90	
	5	10	
		90	

Fractures that intersect at a high angle of 90 degrees take on the shape of deformed parallelepiped cuboids. In low roughness models, these fractures have a straightened and mostly consistent aperture. However, in high roughness models, the fractures are twisted with variable aperture.

The 10 ° fracture intersection are flat and wider than the 90° models with many pieces, which increases with the increase of roughness degree. The white spots inside the low-angle intersection models are the contact area that was initially introduced in the single fracture models.

While the volume of intersections is only a small part of the overall volume (Table. 4) of the X-shaped fracture network, the pressure gradient is noticeably affected by the angle of the intersection in models that are rougher with low apertures.

Table 4. Summary of the intersection volume and the fracture volume at different apertures, roughness and angles.

e (Aperture)	JRC	Angle	Intersection volume mm^3	Fracture volume mm^3	fraction %
1 mm	15	10°	180.3023	3046.9132	5.917530
		90°	32.8641	3037.0192	1.082116
	10	10°	170.91	2836.1764	6.026071
		90°	28.8575	2975.747	0.969757
	5	10°	140.1884	2861.7812	4.898641
		90°	21.6341	2967.0265	0.729151
0.7 mm	15	10°	106.6376	2135.906	4.9926
		90°	18.0827	2136.9073	0.846208
	10	10°	97.2988	1996.1406	4.874346
		90°	15.0143	2092.7263	0.717452
	5	10°	65.8085	2021.0987	3.256076
		90°	10.2832	2090.7707	0.491838

To investigate the effect of the intersection angle on the fluid flow, a series of x-shaped parallel plates fractures simulation has been conducted (Fig.16)

The parallel plate models show a linear relationship between the flow rate and pressure drop for the 10° and 90° models across different aperture sizes (1mm and 0.7mm).

The 90° models show more pressure drop compared with the 10° cases. However, this is more accentuated in the low-aperture models, which means that for the parallel-plate models, the angle affects the fluid flow at the low aperture.

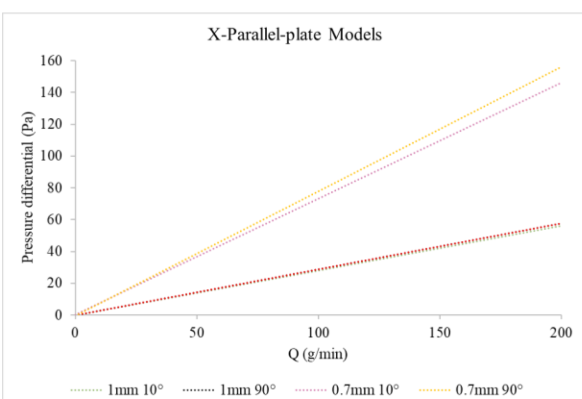


Fig.16. Pressure gradient ΔP against flow rate Q for the parallel plate models (JRC=0) at different apertures and intersection angles.

As the roughness degree increases and the aperture decreases, the impact of the intersection angle on the nonlinearity of fluid flow behavior becomes increasingly prominent. In contrast, for large aperture models, Figure 15 demonstrates that the intersection angle has low effects, regardless of the degree of roughness.

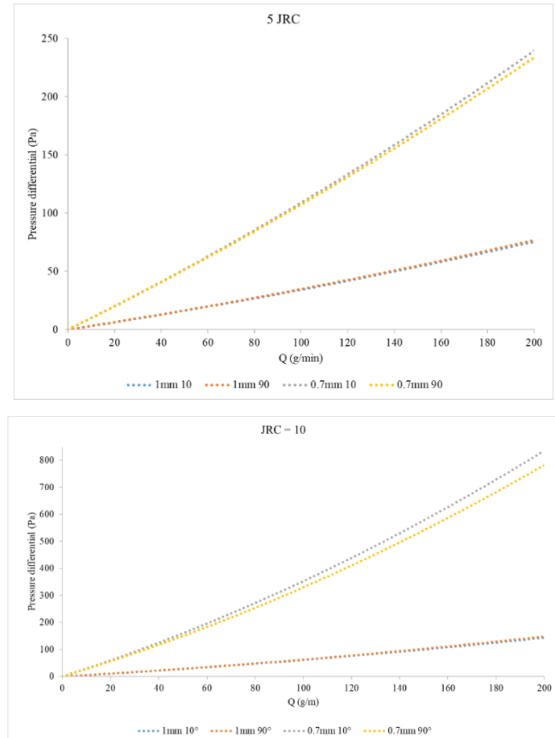


Fig.17. Pressure gradient ΔP against flow rate Q for the 5 (top) and 10 (bottom) JRC roughness models at different intersection angles and apertures.

To investigate the evolution of fluid pressure further, the pressure variation along the X direction was quantified. Fig. 18 is an example of the pressure drop along the X (Y=0) direction under the inlet flow rate of 100 g/min (0.7 mm, 10°, and 5 JRC model) that shows a different behavior of the pressure drop after the intersection. This means that even if the two intersected fractures have exactly the same geometrical characteristics, the intersection can affect the pressure drop along the two outlet branches.

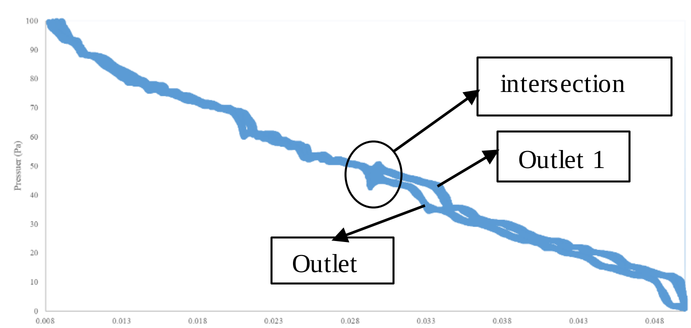


Fig.18 Pressure drop along the x-axis at Y=0 for the model 0.7 mm,90 degrees, and 5 JRC.

This phenomena is not present at the parallel plate models, and the pressure along the tow intersected fractures is the same (Fig.19).

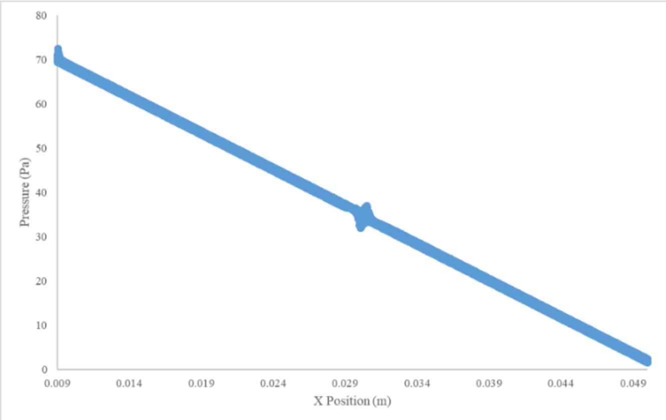


Fig.19 Pressure drop along the x-axis at Y=0 for the model 0.7 mm,90 degrees and JRC=0.

4. CONCLUSION

In this study, a series of 3D x-shaped fracture simulations are developed to investigate the combined effect of geometric properties and relationships between the hydraulic gradient and the flow rate for twelve X-shaped numerical fracture models. These models show different cross angles and surface roughness have been analyzed.

The following conclusions can be obtained:

- The pressure gradient is influenced by various factors, including the intersection angle, roughness degree, and aperture. In particular, the pressure drop tends to be higher in low angle intersection models as the aperture decreases and the degree of roughness increases. This is due to the fact that the intersection angle has a more pronounced impact on the behavior of the fluid flow in such cases. As the aperture decreases, the fluid flow becomes more restricted, and any increase in roughness exacerbates this effect, resulting in a greater pressure drop. Therefore, it is important to consider the geometrical characteristics of the fractures and their intersection when designing or optimizing fluid flow in fractured reservoirs to ensure optimal prediction.

- The intersection angle's impact on fluid flow behavior is minimal for large aperture models, regardless of the degree of roughness. This could be attributed to the fact that larger aperture sizes provide a more significant cross-sectional area for fluid flow, making the impact of the intersection angle less noticeable.

- The intersection geometry plays a vital role in controlling the pressure gradient and fluid flow behavior after the intersection. Even when the intersecting fractures have the same geometrical characteristics, the intersection angle can have a significant influence on the flow behavior. This is because the intersection can alter the direction and velocity of fluid flow, leading to a difference in the pressure gradient between the two outlet branches. Therefore, understanding the impact of intersection and its control over pressure gradients is essential in designing and optimizing the production of fluids from fractured reservoirs. Accurately predicting the behavior of fluid flow through intersections can help identify potential issues and optimize production, resulting in increased efficiency and productivity.

- The fracture intersection geometry is a complex interplay of various factors, including roughness, aperture, shear amount, contact area, and intersection angle. These factors determine the degree of connectivity between the fractures and the resulting flow behavior.

- The intersection volume increases with a larger aperture and decreases with a greater angle of intersection. However, it is difficult to solely link the intersection volume to roughness due to potential contact areas at the junction.

- The intersection geometry becomes intricate at low angles and high roughness, as a result of contact areas and the increasing number of fragments due to the roughness.

- It can be seen that the flow in the rough X-shaped models is clearly nonlinear or non-Darcy for the cases simulated in this study, and the intersection angle has a great impact on the nonlinear behavior.

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